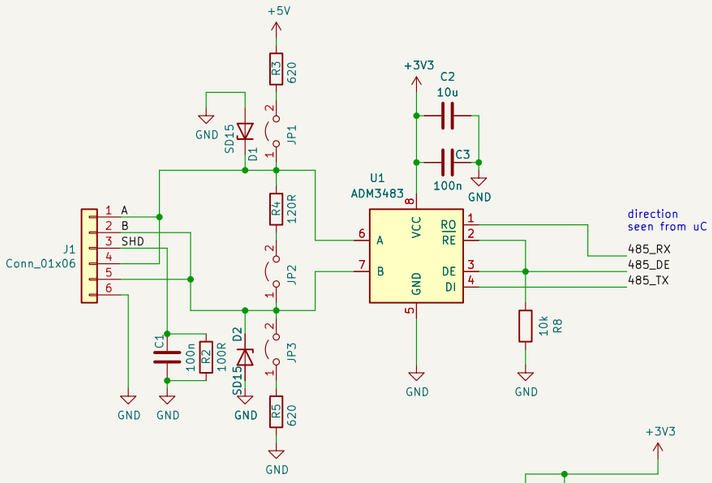
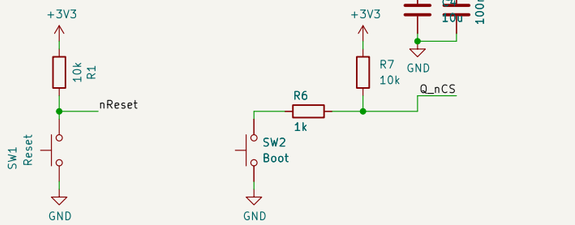
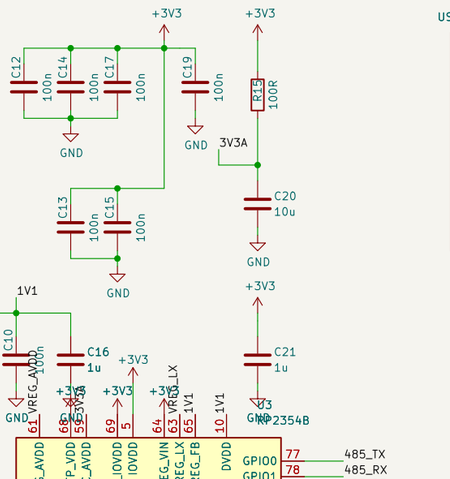
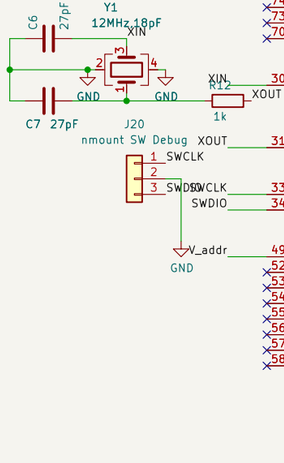
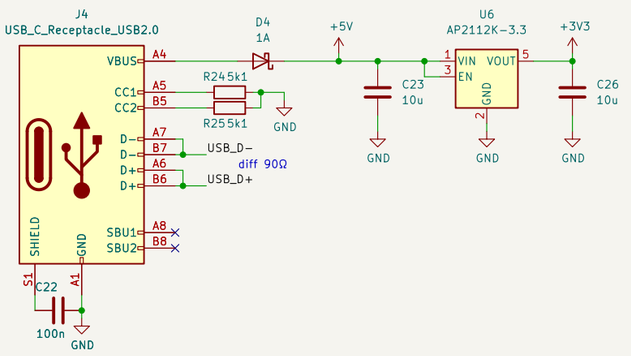
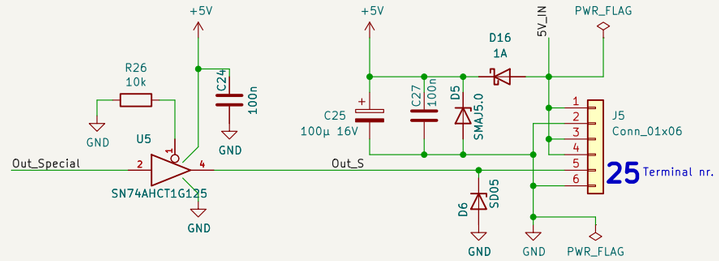


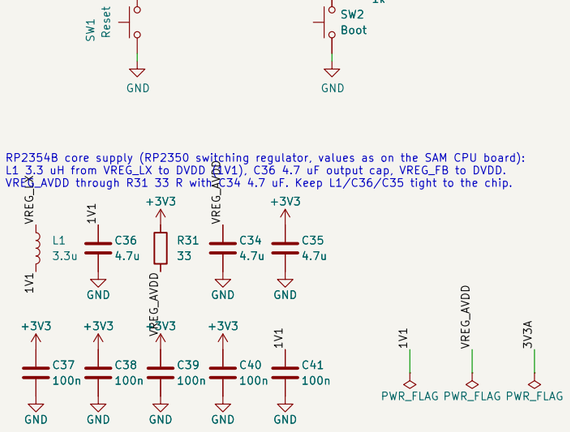










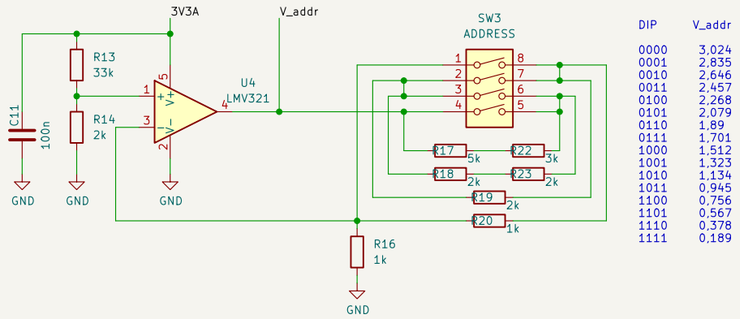


RP2354B core supply (RP2350 switching regulator, values as on the SAM CPU board):
L1 3.3 uH from VREG_LX to DVDD (1V1), C36 4.7 uF output cap, VREG_FB to DVDD.
VREG_AVDD through R31 33 R with C34 4.7 uF. Keep L1/C36/C35 tight to the chip.

SAM_IO, derived from PPUC IO_16_8_1 v1.1.1 (TAPR OHL 1.0, see ../README.md):
- IN16 (16 inputs, J6–J8) and OUT8 (8 high-power outputs, J9–J11, F2, C56) removed.
- MCU is an RP2354B (2 MB internal flash, QFN-80); the RP2040 and the W25Q flash are gone. QSPI pins unused, QSPI_SS
- GPIO3–18 drive the SAM bus through level shifters on sheet 'SAM bus' (J9 to the SAM IO board J1).
- GPIO19–24, 26, 27 = dedicated D17–D24, GPIO30–37 = D9–D16, GPIO38 = coin door memory protect (sheet 'Dedicated
- GPIO25 LED, GPIO29 special output. Board address (SW3) moved to GPIO40/ADC0. SWD header is now J20.

SAM bus
File: sam_bus.kicad_sch

Dedicated switches
File: dedicated_sw.kicad_sch



DIP	V_addr
0000	3,024
0001	2,835
0010	2,646
0011	2,457
0100	2,268
0101	2,079
0110	1,89
0111	1,701
1000	1,512
1001	1,323
1010	1,134
1011	0,945
1100	0,756
1101	0,567
1110	0,378
1111	0,189

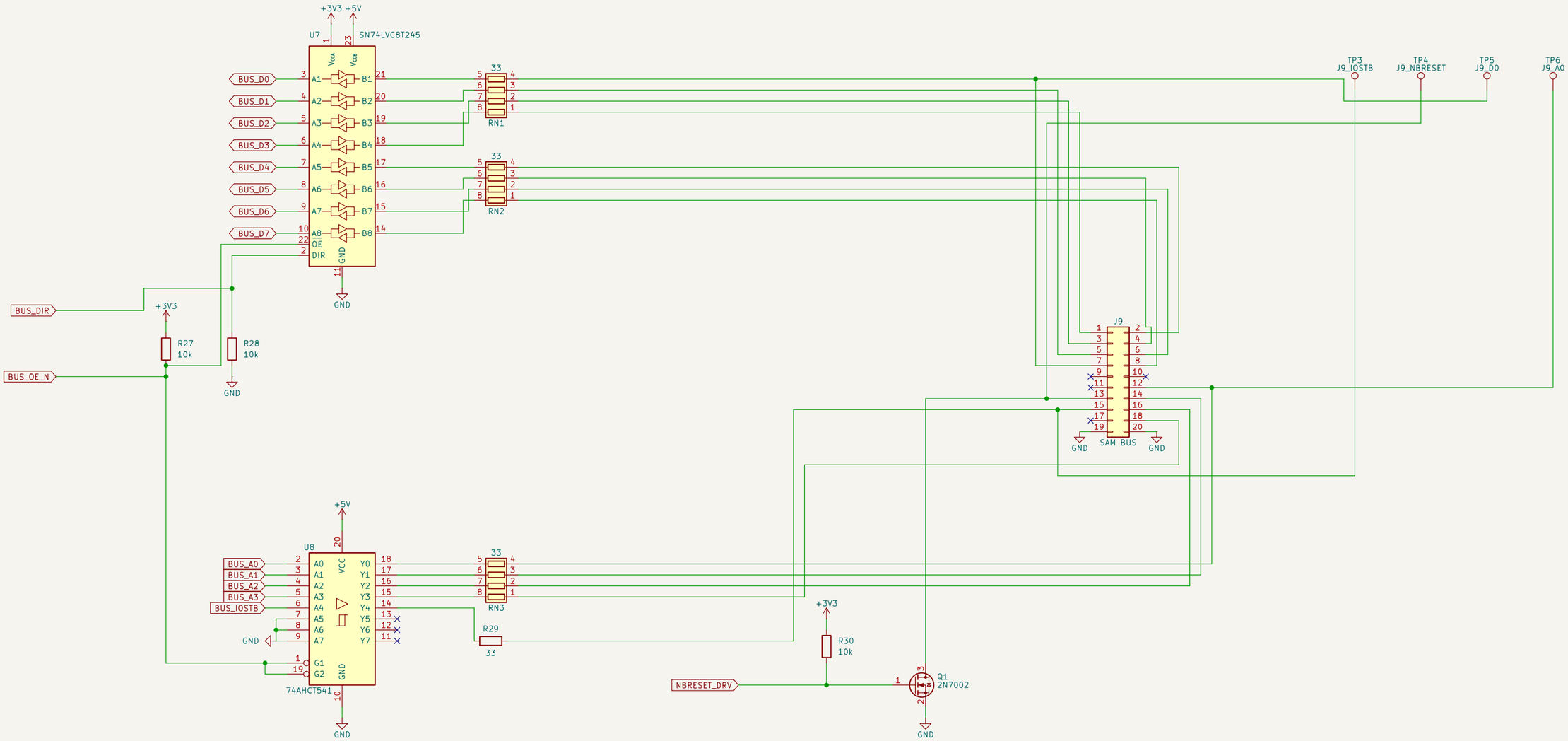
PPUC

Sheet: /
File: sam_io.kicad_sch

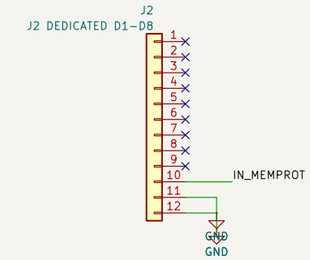
Title: SAM_IO

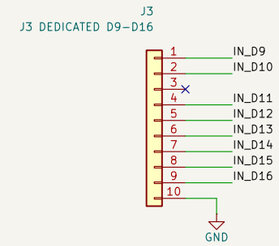
Size: A3	Date: 2026-10-07	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 1/3

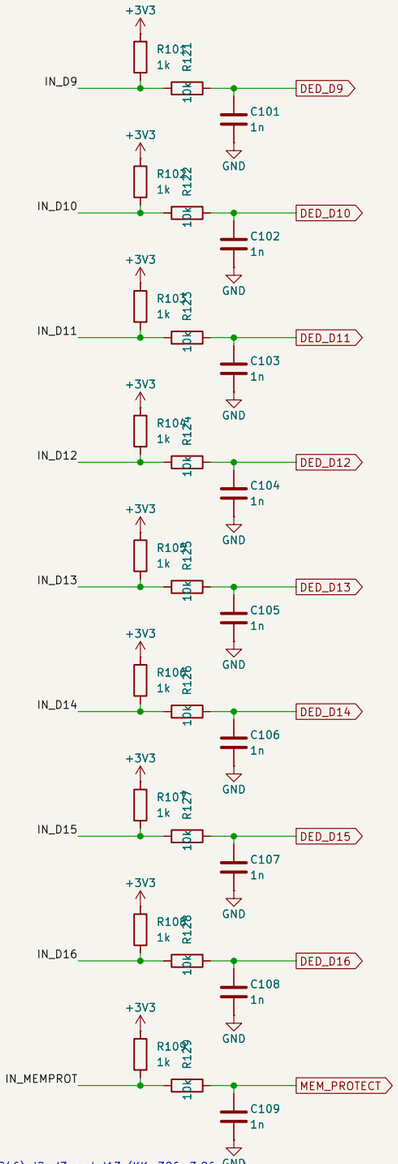
Copy of the SAM CPU replacement board IO bus sheet (Stern-SAM-CPU-PPUC hardware/kicad, ARCHITECTURE.md 3.1-3.2).
J9 has the SAM IO board J1 pinout, pin for pin: a straight 20-way ribbon connects the two.
GPIOs from io-boards src/IODevices/SamBus/SamBusPins.h: D0-D7 GPIO3-10, A0-A3 GPIO11-14, IOSTB 15, DIR 16, OE_N 17, NBRESET_DRV 18.
Both buffers are off at power-up (BUS_OE_N pulled high) and NBRESET is held low until the firmware releases it.
33 R series resistors on the J9 side for the ribbon cable. Unused 74AHCT541 inputs tied low.

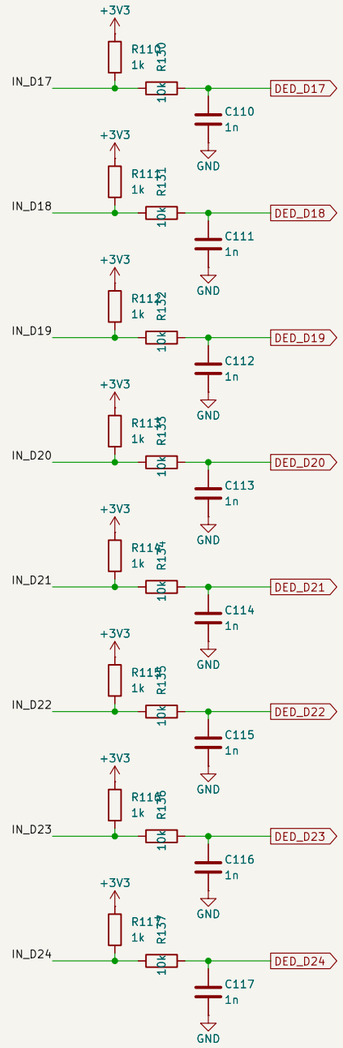


Sheet: /SAM bus/		
File: sam_bus.kicad_sch		
Title: SAM bus: J9 to the SAM IO power driver board J1		
Size: A3	Date: 2026-10-07	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 2/3









SAM dedicated switch inputs, harness compatible with the SAM CPU board (520-5246) J2, J3 and J13 (KK-396, 3.96 mm).
Pin order follows the SAM CPU board schematic in this repository (hardware/kicad/sam_cpu, sw_dedicated_1/2): check it with a meter.
Routed: J3 D9–D16 (flipper buttons and EOS), J13 D17–D24 (tilt, slam, ticket notch, D20, BACK, MINUS, PLUS, SELECT)
and J2 pin 10 (coin door memory protect). Not routed: J2 D1–D8 (coin slots), key pins.
Per input: 1 k pull-up to +3.3 V (about 3.3 mA wetting current, the original uses 1.5 k to 4.5 V),
10 k series and 1 nF (10 us) to the RP2354B. Switch closed = low. No 5 V reaches the MCU.
Tron LE names (rom_data/states/dedicated_switches.csv): D9 L flipper, D10 L EOS, D11 R flipper, D12 R EOS, D13 UL flipper,
D17 tilt, D18 slam, D19 ticket notch, D21 BACK, D22 MINUS, D23 PLUS, D24 SELECT. D14–D16 and D20 vary per game.

Sheet: /Dedicated switches/
File: dedicated_sw.kicad_sch

Title: Dedicated switches: SAM harness J2, J3, J13

Size: A3	Date: 2026-10-17	**Rev: 0.2**
KiCad E.D.A. 10.0.6		Id: 3/3